

# Proofs About Sets & Quantification

Robert Y. Lewis

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# Overview

1 Proofs about Sets, Proof by Cases (1.7)

2 Predicate Formulas (3.6)

# Set equality

When are two sets equal?

If  $A$  and  $B$  are sets,  $A = B$  if and only if  $\forall x, x \in A \leftrightarrow x \in B$ .

Equivalently:  $(\forall x, x \in A \rightarrow x \in B) \wedge (\forall x, x \in B \rightarrow x \in A)$ .

Equivalently:  $A \subseteq B \wedge B \subseteq A$ .

This suggests a proof technique for proving set equality: the *set-element method*.

## Proving set equalities: set-element method

Theorem: For any sets  $A$  and  $B$  of elements in universe  $U$ ,  $\overline{A \cap B} = \bar{A} \cup \bar{B}$ .

(DeMorgan's Law again! Now, connects intersection and union instead of  $\wedge$  and  $\vee$ .)

Proof. We proceed by the set element method. First, we show  $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$ . Suppose  $x \in \overline{A \cap B}$ . This means that  $x$  is *not* in both  $A$  and  $B$ . Written as a formula:

$\neg(x \in A \wedge x \in B)$ . By De Morgan's law, this is equivalent to  $\neg(x \in A) \vee \neg(x \in B)$ . So  $x \in \bar{A} \vee x \in \bar{B}$ , as desired.

Now we show  $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$ . Suppose  $x \in \bar{A} \cup \bar{B}$ . This means that  $x \in \bar{A} \vee x \in \bar{B}$ . Equivalently,  $\neg(x \in A) \vee \neg(x \in B)$ . By De Morgan again, this is equivalent to  $\neg(x \in A \wedge x \in B)$ , so  $x \in \overline{A \cap B}$  as desired.

## Fact about groups of people

Any two people have either met or not.

Given a set of people  $G$ , if all pairs of people in  $G$  have met, we'll call it a *club*. If no two people in  $G$  have met, we'll call them *strangers*.

**Theorem.** Every collection of 6 people includes a club of 3 people or a group of 3 strangers.

Does that seem true? Try some examples on the board.

## Proof (Part 1)

The proof is by case analysis. Let  $x$  denote one of the six people. Let  $R = G - \{x\}$  be the rest. There are two cases:

- 1 Among  $R$ , at least 3 have met  $x$ .
- 2 Among  $R$ , at least 3 have *not* met  $x$ .

At least one of these cases must hold. Since  $|R|$  is odd, either more than half in  $R$  know  $x$  or less than half in  $R$  know  $x$  (and therefore more than half do not know  $x$ ).

Case 1: At least 3 have met  $x$ . Let  $J \subseteq R$  be those individuals. Two subcases:

- 1.1 No pair in  $J$  have met each other. So,  $J$  is a group of at least 3 strangers and the theorem holds in this subcase.
- 1.2 Some pair in  $J$  have met each other. That pair and  $x$  are a club of 3 people and the theorem holds in this subcase, too.

That covers Case 1!

## Proof (Part 2)

Case 2: At least 3 have not met  $x$ . Let  $J \subseteq R$  be those individuals. Two subcases:

- 2.1 Every pair in  $J$  have met each other. So,  $R$  is a club of at least size 3 and the theorem holds in this subcase.
- 2.2 Some pair in  $J$  haven't met each other. That pair and  $x$  are a group of strangers of 3 people and the theorem holds in this subcase, too.

That covers Case 2! It's kind of the inverse-video version of Case 1.

Since we showed that only these two cases can occur and the theorem holds in both, the theorem *always* holds.

## Mixing quantifiers

**Theorem** (sparse squares): There's a perfect square arbitrarily far from its closest perfect square.

Clear? Maybe a tad vague. True? How do we say this in logic?

$$\forall d : \mathbb{N}, \exists i : \mathbb{N}, \forall j : \mathbb{N}, \\ (i \text{ is a perfect square}) \wedge (|i - j| \leq d \rightarrow \neg(j \text{ is a perfect square})).$$

The expressions nest inside each other. The order matters.

You can think of it like a little game. I'm claiming that you can pick any  $d$  you want. I'll then pick an  $i$  that's a perfect square AND no matter what  $j$  you pick that is within  $d$  values of  $i$ ,  $j$  won't be a perfect square.

So, what's my winning strategy?

## Any ambiguity is too many

“If you can identify any bird, you’ve got talent.”

- 1 If  $\exists b$ , you can identify  $b$ , then you’ve got talent.
- 2 If  $\forall b$ , you can identify  $b$ , then you’ve got talent.

“...statistics show that, in the UK, someone brews a cup of tea every second.”

- 1  $\forall t, \exists p, p$  brews a cup of tea at second  $t$

“That person’s name is Nigel.”

- 2  $\exists p, \forall t, p$  brews a cup of tea at second  $t$

## DeMorgan returns: Negating quantifiers

These two statements are equivalent:

- Not everyone likes coffee.
- There's someone who doesn't like coffee.

$\neg \forall x, P(x)$  is equivalent to  $\exists x, \neg P(x)$ .